

Running Linux on Windows

One of the many reasons to run Linux on Windows would be to get the best of both the worlds. There are several methods for running Linux on Windows. The authors recommend Linux Integration Services, which has been made available by Microsoft. Read on to learn how to use LIS



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Recent releases from Microsoft—Windows Server 2012 R2 and System Center 2012 R2—come with the ability to effectively host Linux and associated open source applications along with great management capabilities. Some of the improvements in Windows Server include faster live migration, dynamic memory support that can be used depending on workload, as well as the capability to dynamically resize VHDs and VHDXs. On the other hand, System Center provides a single interface for managing Windows, ESXi and Linux operating systems.

Microsoft's new feature from Windows Server 2012 R2 and System Center 2012 R2, called Data Center Abstraction Layer (DAL), based on CIM and WS-Man standards, has common management abstracts for managing all the resources of a data centre (including the physical and virtual environ-

ments). To support DAL, Microsoft has contributed Open Management Infrastructure (OMI) along with other providers to effectively manage Linux. OMI is well known for its smaller disk footprint.

This article will tell you how to use Windows Server 2012 R2 Hyper-V and Windows Hyper-V Server 2012 R2, and why installing Linux Integration Services is a best practice while running Linux as a guest OS.

Different ways of running Linux on Windows

Linux can be run on Windows in the following ways.

Virtual machines. Virtual machines (VMs) are some of the best ways of running any operating system. The advantages of a VM include better hardware utilisation, the ability to run multiple OSs and to install and run any application. On a Windows machine, one can install third-party virtualisa-

tion software like Virtual Box, VMware Player or Windows Hyper-V. Note that one can't install multiple virtualisation software on the same system.

The dual boot option. The dual boot option allows more than one operating system to reside on a single computer. In a dual boot, the Windows OS is stored on one volume/disk and the Linux OS on another volume/disk (assuming it is rarely used). The best use case would be to test a new OS without switching to it completely. This method is not the recommended way of running Linux alongside Windows, as the latter facilitates only partition-specific installation or on a logical partition (within an extended partition). The Microsoft recommendation is to install Windows on a primary partition. To dual boot a Windows machine, please refer to the link <https://help.ubuntu.com/community/Installation/FromUSBstick>

Live USB and live CDs. A live USB is a USB Flash drive or USB external hard drive containing the full operating system. It is the preferred way of running Linux when compared to a live CD, as data contained in the booting device can be changed and additional data can be stored in the USB. Refer to the link <https://help.ubuntu.com/community/Installation/FromUSBstick> to boot from a live USB.

Installing Ubuntu via Wubi. Windows-based Ubuntu Installer (Wubi) is a software installer for Ubuntu. Wubi creates a special file on your Windows partition and uses that file as your Ubuntu drive without the need for a separate partition. This means that you can install Ubuntu and use it without any partitioning, and you can uninstall Ubuntu from the Windows Control